Lecture 6: Power MOSFET and IGBT

EEE 339 Power Electronics · Previous-year question solutions · Source: *class slides/Lecture 06 Power MOSFET and IGBT.pdf* (Rashid ch. 4)

Questions from this lecture in the 2021 to 2024 finals:

Year	Question	Marks
2023 Q3(b)	Draw the cross-section of an IGBT showing the layers; differentiate PT and NPT IGBTs	≈ 6
2022 Q6(d)	Briefly describe the half-cell model of an IGBT with appropriate figure	7
2022 Q6(b)	Write the advantages of using IGBT as a switch	4
2021 Q3(d)	With diagram, explain the pinch-off condition in a power MOSFET	4
2022 Q4(a), 2021 Q1(a)	BJT vs MOSFET; BJT vs MOSFET vs IGBT comparison	5, 7

The IGBT was asked in 2022 and 2023, so it is the most likely topic from this lecture again.

Q1 · 2023 Q3(b): IGBT cross-section; PT vs NPT IGBT

Q. Draw the cross-section of an IGBT showing the layers. Differentiate between punch-through and non-punch-through IGBTs.

(a) Cross-section

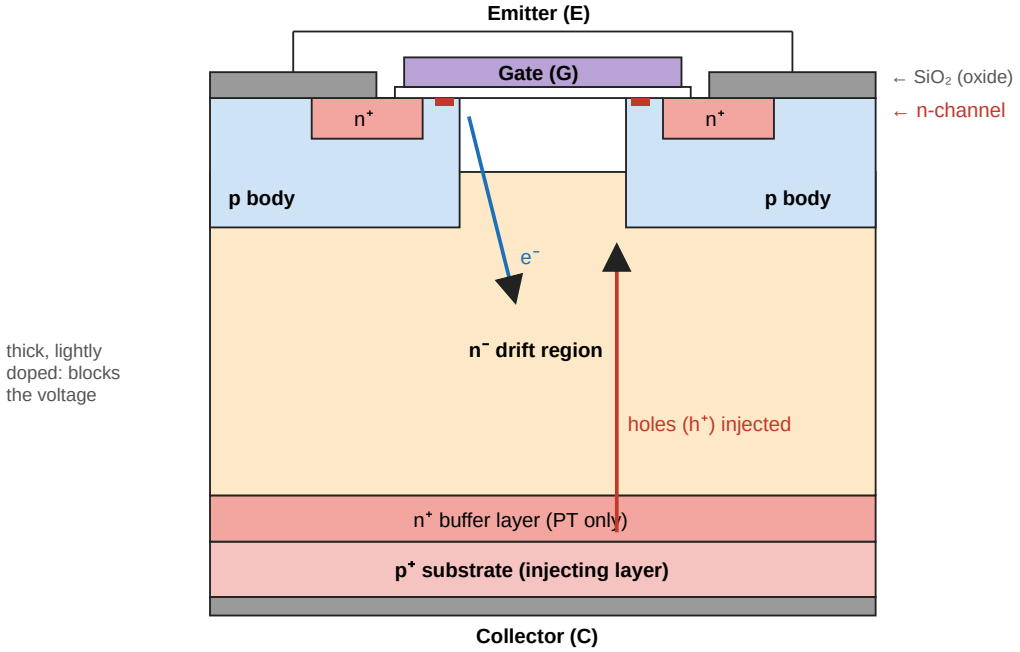


Fig. 1: Cross-section of one IGBT cell (punch-through type). The NPT type has no n^+ buffer layer.

Layers from collector to emitter:

1. **p^+ substrate (collector):** the injecting layer. It injects holes into the drift region when the device is on. This is what makes the IGBT different from a MOSFET (a MOSFET has n^+ here).
2. **n^+ buffer layer** (PT only): stops the depletion region from reaching the p^+ layer.
3. **n^- drift region:** thick and lightly doped; it supports the blocking voltage.
4. **p body:** the channel is formed in this layer under the gate.

5. **n^+ emitter** regions inside the p body, contacted by the emitter metal (the metal also shorts n^+ to the p body).
6. **Gate:** polysilicon over a thin **SiO_2** layer, so the gate is insulated (very high input impedance).

Operation in one line: V_GE > V_th creates an n-channel in the p body, like a MOSFET. Electrons flow from the n^+ emitter through the channel into the drift region; the p^+ layer then injects holes. The drift region is flooded with carriers (**conductivity modulation**), so the on-state drop is low.

(b) PT vs NPT IGBT

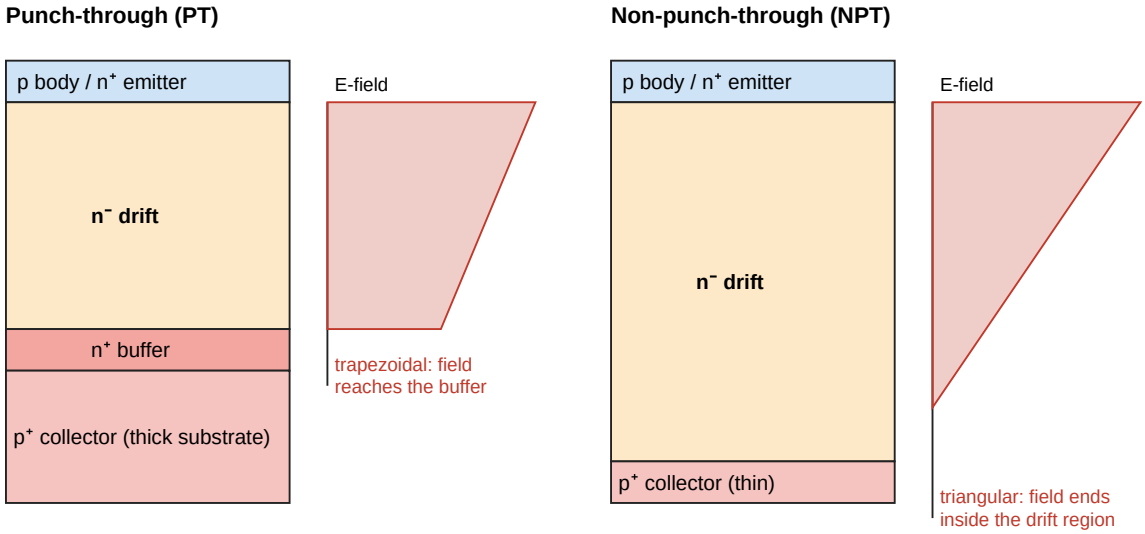


Fig. 2: Layer stacks and electric-field shapes of PT and NPT IGBTs in the blocking state.

Feature	Punch-through (PT)	Non-punch-through (NPT)
n^+ buffer layer	**Yes**, between p^+ and n^-	**No**
Field in blocking state	Depletion "punches through" the drift region to the buffer (trapezoidal field)	Depletion ends inside the drift region (triangular field)
Drift region	Thinner	Thicker
p^+ collector	Thick, heavily doped substrate (epitaxial wafer)	Thin, lightly doped layer on the back of the wafer
On-state drop V_CE(sat)	Lower	Slightly higher
Turn-off / tail current	Faster (with lifetime control); switching losses rise with temperature	Longer tail but losses nearly independent of temperature
Reverse blocking	Very low (asymmetric)	Can block reverse voltage (symmetric)
Temperature coefficient of V_CE(sat)	Negative at normal currents: harder to parallel	**Positive**: easy to parallel
Ruggedness	Lower short-circuit capability	More rugged (better short-circuit capability)

Q2 · 2022 Q6(d): Half-cell model of an IGBT [7]

Q. Briefly describe the half-cell model of an IGBT with appropriate figure.

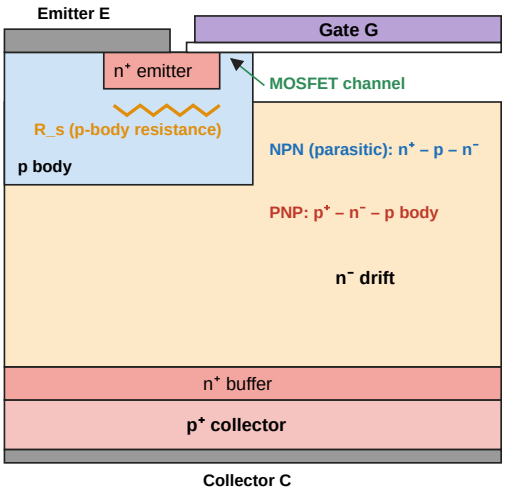


Fig. 3a: Half cell with the parasitic devices.

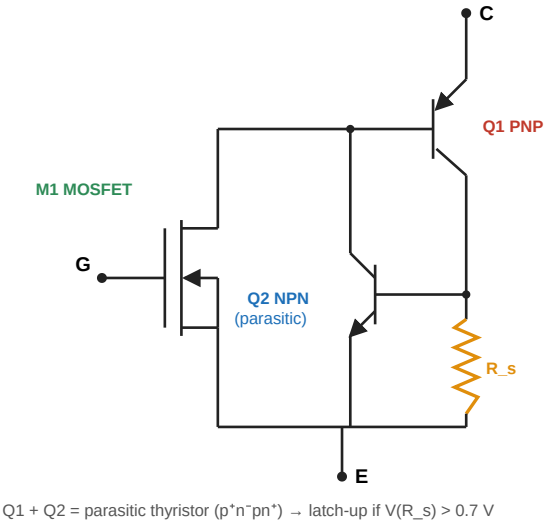


Fig. 3b: Equivalent circuit of the half cell.

An IGBT chip is made of thousands of identical cells. Because each cell is symmetric, **half a cell** is enough to model the device. The half cell contains:

1. **MOSFET M1**: n^+ emitter, p body (channel) and n^- drift, controlled by the insulated gate.
2. **PNP transistor Q1**: p^+ collector (emitter of Q1), n^- drift (base of Q1), p body (collector of Q1). This is the main bipolar path.
3. **Parasitic NPN transistor Q2**: n^+ emitter, p body (base), n^- drift (collector).
4. **R_s**: the lateral resistance of the p body under the n^+ emitter, between the base of Q2 and the emitter contact.

Normal operation:

- A gate voltage above V_th creates the channel. The MOSFET's electron current flows into the n^- drift, which is the **base current of the PNP**.
- The PNP turns on and the p^+ layer injects holes: conductivity modulation gives a low V_CE(sat).
- So the IGBT behaves like a **MOSFET driving a PNP transistor** (the slide calls it a Darlington of an IGFET and a BJT): $I_C = I_{MOS} + \beta_{PNP} \cdot I_{MOS} = (1 + \beta_{PNP}) \cdot I_{MOS}$.
- The holes collected by the p body flow to the emitter contact through R_s.

Latch-up:

- If the hole current is too large, the drop across R_s exceeds about 0.7 V and forward-biases the n^+–p junction, so the **parasitic NPN turns on**.
- Q1 and Q2 then form a **parasitic thyristor** ($p^+n^-pn^+$) that latches on, and the gate **loses control**; the device can be destroyed.
- **Prevention:** keep R_s small (heavy p^+ doping under the emitter, short n^+ regions), short the n^+ and p body with the emitter metal, and limit the current.

Q3 · 2022 Q6(b): Advantages of IGBT as a switch [4]

1. **Voltage-controlled** insulated (MOS) gate: very high input impedance, negligible gate current, simple and cheap drive circuit.
2. **Low on-state voltage drop** (like a BJT) because of conductivity modulation, so low conduction loss, much lower than a high-voltage MOSFET.
3. **High voltage and current ratings** (up to several kV and kA) with high current density, so a smaller chip.
4. **Faster switching than a BJT**; good for PWM inverters and drives (about 20–50 kHz).
5. Turned on and off by the gate: **no commutation circuit** needed (unlike an SCR).
6. Wide safe operating area; less prone to second breakdown than a BJT.
7. NPT IGBTs have a positive temperature coefficient, so they are **easy to parallel**.

If the question also asks for disadvantages (slide): slower turn-off than a BJT or MOSFET because of the **tail current**; faster turn-off is possible only with a higher V_CE(sat); latch-up risk.

Q4 · 2021 Q3(d): Pinch-off condition in a power MOSFET [4]

Q. With diagram, explain the pinch-off condition in a power MOSFET.

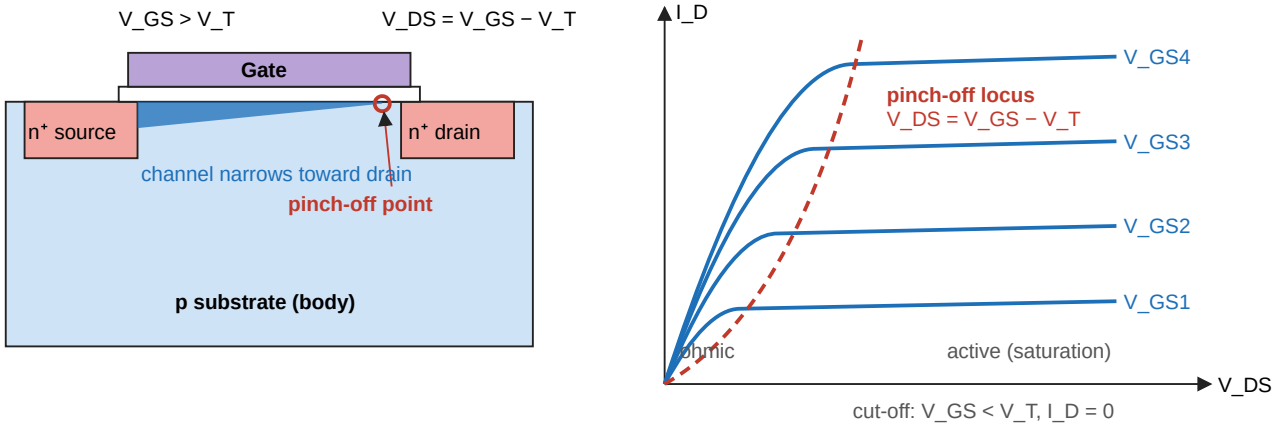


Fig. 4: Left: channel pinch-off at the drain end. Right: output characteristics; the pinch-off locus separates the ohmic and active regions.

1. **Channel formation:** for an n-channel enhancement MOSFET, a gate-source voltage V_GS above the threshold V_T induces an n-type inversion channel between source and drain.
2. **Ohmic region:** for a small V_DS the channel has the same depth along its length and behaves like a resistor (R_DS(on)); I_D rises almost linearly with V_DS.
3. **Channel narrowing:** as V_DS increases, the gate-to-channel voltage at the drain end, V_GD = V_GS − V_DS, decreases, so the channel becomes thinner near the drain.
4. **Pinch-off:** when V_GD = V_T, that is

   $$V_{DS} = V_{GS} - V_T$$

   the channel depth at the drain end becomes zero. This is the **pinch-off condition**.
5. **Beyond pinch-off** (V_DS > V_GS − V_T): the extra voltage is dropped across the small depletion region near the drain; the current stops increasing and stays at $I_D \approx K(V_{GS} - V_T)^2$. The MOSFET is in the **active (saturation) region**, where I_D is controlled by V_GS only.
6. The locus of pinch-off points, $\mathbf{I_D = K \cdot V_{DS}^2}$, is the boundary between the ohmic and active regions (dashed line in Fig. 4).

For switching, a power MOSFET is used in the ohmic region (ON) and cut-off (OFF); the active region is crossed only during transitions.

Q5 · 2022 Q4(a) and 2021 Q1(a): Comparisons

These are answered in full in your **Lecture 5 notes** (pages 8 to 9): BJT vs MOSFET (10-row table) and BJT vs MOSFET vs IGBT (power handling, input impedance, switching speed, on-state resistance, blocking voltage). Short version to remember:

Parameter	BJT	MOSFET	IGBT
Control / input impedance	Current / low	Voltage / very high	Voltage / very high
Speed	Slow	Fastest	Medium
On-state drop	Low	High at high voltage	Low
Power / voltage	Medium	Low-medium	High
Second breakdown	Yes	No	Less prone

Likely new questions from this lecture (short answers)

Predicted **Q. Write the features and disadvantages of a power MOSFET.**

Features (slide): low input current; voltage-controlled; high switching speed; no second breakdown; high input impedance; unipolar; no commutation circuit needed; small size; small holding power.
Disadvantages (slide): high on-state voltage, so high conduction loss compared with a BJT; easily damaged by static electricity, so it needs careful handling; asymmetric blocking: blocks high forward voltage but not high reverse voltage.

Predicted **Q. Why is a vertical (VDMOS) structure used for power MOSFETs?**

In a lateral MOSFET both the current and the breakdown voltage depend on the channel dimensions, so it is inefficient. In the vertical, double-diffused MOSFET the source is on top and the drain at the bottom, so current flows vertically. The **voltage rating depends on the doping and thickness of the n^- layer**, while the **current rating depends on the channel width**. The device can therefore handle high voltage and high current without becoming large. "Double-diffused" refers to the p and n^+ wells made by two diffusions.

Predicted **Q. Why was the IGBT developed? / Explain the turn-off of a power MOSFET.**

Why IGBT: a BJT needs a large base current (even as a Darlington); a MOSFET needs almost no gate current but has a high on-state drop at high current. The IGBT combines the MOSFET's insulated gate with the BJT's low on-state drop.
MOSFET turn-off (slide): t5: v_GS falls, C_GS discharges and C_GD charges; i_D and v_DS unchanged. t6: v_DS rises to V_CC while v_GS stays constant (Miller plateau); then the diode turns on and i_D starts to fall. t7: v_GS falls to V_T and i_D falls to zero (cut-off). t6 and t7 have both i_D and v_DS present, so they cause the switching loss.